Technical Dossier and Research Compendium: The Structural Biology Revolution through the Work of Jane Shelby Richardson

This comprehensive research dossier serves as a high-signal repository of historical, technical, and biographical data concerning Jane Shelby Richardson. Prepared for an introductory biochemistry audience, the report avoids finished narrative prose in favor of exhaustive evidence compilation. It prioritizes the evolution of molecular visualization and the establishment of protein taxonomy as seen through primary sources, oral histories, and archival finding aids.

A) ANCHOR SCENES (3–6)

SCENE #1: The Sputnik Observation and the Westinghouse Science Talent Search

Summary: In 1957 and 1958, a seventeen-year-old Jane Shelby Richardson performed a series of independent astronomical observations and mathematical calculations to determine the orbit of the Sputnik I satellite. This event established her trajectory as a rigorous observer of natural phenomena and a master of spatial data, long before she entered the field of biochemistry.[1] **Date/Place**: October 1957 – 1958; Teaneck, New Jersey, and Washington, D.C..[1] **People**: Jane Shelby Richardson; her "astronomically inclined" high school friends; her father (an electrical engineer) and mother (an English teacher).[3] **Concrete details**:

- **The Hardware**: Richardson conducted naked-eye observations combined with data from a hand-built 6-inch telescope mirror that she had ground and polished herself.[4]
- **The Process**: She tracked the satellite across successive nights to plot its path, translating these points into a predicted orbit using physics and math.[3]
- **The Setting**: Cold nights in Teaneck, New Jersey, counting meteors and watching for the man-made light of Sputnik.[4]
- **The Achievement**: She traveled to Washington, D.C., for the Westinghouse Science Talent Search (STS), where she was awarded third place out of the entire nation.[1] **Why it matters**: This scene establishes her early commitment to "observation and instrumentation".[5] It highlights her ability to extract high-level patterns from raw, complex data—a skill that would later define her work in protein taxonomy.[1] It also provides the "unconventional" start to her career, moving from physics and astronomy to philosophy, then finally to biology.[1] **Evidence quality**: High. Documented in institutional records of

the Society for Science and multiple oral history profiles.[1] **Myth risk**: Low. The details regarding the orbit calculation and the placement in the STS are factual and widely corroborated by the awarding body.[2] **Sources**:.[1]

SCENE #2: The MIT Nuclease Struggle and the Tygon Tubing Experiment

Summary: While working as a technician in Al Cotton's lab at MIT, Jane and David Richardson struggled to visualize the complex 3D "fold" of the Staphylococcal nuclease. This led to a tactile, DIY experiment suggested by Chris Anfinsen to reveal the backbone.[7] **Date/Place**: 1964–1969; Massachusetts Institute of Technology (MIT), Cambridge, MA.[1] **People**: Jane Richardson, David Richardson, Al Cotton (Lab PI), Chris Anfinsen (Protein folding pioneer).[7] **Concrete details**:

- **The Problem**: The team had determined the structure of Staphylococcal nuclease—only the 10th protein structure ever solved—but the 3D brass model was a dense forest of wire that obscured the overall path of the amino acid chain.[1]
- **The Solution**: Chris Anfinsen suggested tying 1/4-inch Tygon tubing along the backbone of the brass model. They filled the tubing with fluorescent dye and viewed it under UV light.[7]
- **The Result**: Under the UV glow, the specific atomic details vanished, and the "worm-like" path of the backbone emerged as a glowing, continuous line.[7]
- **The Drawing**: Jane made a primitive "worm" drawing from that UV-lit image, which served as the precursor to the ribbon drawings.[7] **Why it matters**: This was the literal "lightbulb moment" for protein visualization. It taught Richardson that "any simplification must omit information," but that removing the "forest" of atoms was necessary to see the "trees" of the protein's architecture.[7] **Evidence quality**: Very High. Detailed in the Richardsons' own retrospective reviews and scientific papers.[7] **Myth risk**: Low to Medium. The detail of David tapping the diffractometer with a hammer for calibration is a vivid lab-culture anecdote, but it is documented in technical histories.[12] **Sources**:.[1]

SCENE #3: The "Greek Key" Realization at the Regional Meeting

Summary: During a chance encounter with crystallographer David Davies, Jane and David Richardson compared the models of two seemingly unrelated proteins—immunoglobulin and superoxide dismutase—and realized they shared a specific, recurring topological motif.[13] **Date/Place**: Mid-1970s (approx. 1975–1976); A regional crystallography meeting.[13] **People**: Jane Richardson, David Richardson, David Davies (NIH molecular biologist).[13] **Concrete details**:

- **The Artifacts**: Davies was carrying a wire model of an immunoglobulin fragment; the Richardsons had their model of Cu,Zn superoxide dismutase (SOD).[13]
- **The Comparison**: They sat down to compare the structures closely and noticed that

both proteins, despite vastly different functions, used the same crisscrossing pattern of β -strands.[13]

- **The Visual Analogy**: They named the pattern the "Greek key fold" because of its similarity to the "meander" designs common on ancient Greek vases, jewelry, and placemats.[13] **Why it matters**: This marks the birth of structural bioinformatics. It shifted the field from viewing proteins as unique "frozen accidents" to recognizing finite "folding preferences".[13] **Evidence quality**: High. Documented in the 1977 *Nature* paper and retrospective interviews.[1] **Myth risk**: Low. The "placemat" and "vase" analogies are widely cited by Richardson herself.[14] **Sources**:.[1]

SCENE #4: The Year of the 75 Proteins (The 1981 Manifesto)

Summary: Commissioned by Chris Anfinsen to write a review for *Advances in Protein Chemistry*, Jane Richardson spent an entire year hand-drawing every known protein structure to create a uniform "taxonomy" of folds.[7] **Date/Place**: 1980–1981; Richardson Lab, Duke University.[1] **People**: Jane Richardson, Chris Anfinsen (Editor of the series).[7] **Concrete details**:

- **The Process**: Drawing on top of C_α -trace stereo prints using colored pencils and pastels.[4]
- **The Technique**: To learn how to draw twisted β -strands, she held up a twisted leather belt to her eyes to study how the edges and sides should curve in 3D.[7]
- **The Scope**: She drew the 75 distinct protein domain structures that had been solved globally by that time.[1]
- **The Inspiration**: She cited MC Escher's "rind" drawings of human heads as a stylistic influence for showing surfaces that are both inside and outside.[7] **Why it matters**: This resulted in the 1981 paper "The Anatomy and Taxonomy of Protein Structure," the most enduring visual language in protein science.[3] **Evidence quality**: High. The paper itself is the primary source, and her process is extensively documented in oral histories.[1] **Myth risk**: Low. The count of "75 proteins" is a hard historical metric.[7] **Sources**:.[1]

SCENE #5: The 2020 SARS-CoV-2 Validation Surge

Summary: Decades after her first drawings, Jane Richardson used her lab's modern software, MolProbity, to serve as a "rigorous peer review" for the flood of new SARS-CoV-2 protein structures, ensuring accurate models for vaccine development.[20] **Date/Place**: 2020–2021; Richardson Lab, Duke University (Digital/Remote).[20] **People**: Jane Richardson, David Richardson, the Coronavirus Structural Task Force.[22] **Concrete details**:

- **The Crisis**: Rapid release of SARS-CoV-2 protein structures often contained "peptide flips" or sequence misalignments.[22]
- **The Tools**: Using "CaBLAM" (C-alpha Based Low-resolution Annotation Method) and

all-atom contact analysis to flag overlaps.[22]

- **The Outcome**: Richardson's team worked "behind the scenes" to get corrections updated in the Protein Data Bank (PDB) by the original depositors.[22] **Why it matters**: Connects her early visual intuition to modern computational validation, proving that accurate representation is a prerequisite for drug design.[21] **Evidence quality**: High. Peer-reviewed articles in *Proteins* document these specific corrections.[22] **Myth risk**: Low. Data-driven technical process. **Sources**:.[20]

B) DIRECT QUOTES FROM THE SCIENTIST (Target 3–7)

1. **On the utility of her non-science degree**: "I think my philosophy background is actually useful, mainly in questioning absolutely everything." *Context/Date*: Frequently cited in retrospective interviews regarding her transition from philosophy to biophysics; reflects her Swarthmore B.A. training.[4]
2. **On the subjectivity of visualization**: "It's also kind of scary! If I didn't do it quite right, they're all stuck with my misconceptions!... a whole generation of scientists see protein structures through my eyes." *Context/Date*: Commenting on the universal adoption of the Richardson diagram after 1981.[11]
3. **On the danger of simplification**: "Such a drawing has inherent dangers... any simplification must omit information... But even if one is led to miss alternative interpretations something worthwhile is gained, because understanding one conceptualization of a structure is much better than not understanding it at all." *Context/Date*: Found in the introductory sections of her seminal 1981 review paper on protein taxonomy.[11]
4. **On the ethos of her lab**: "Share, collaborate, and give away ideas and results freely." *Context/Date*: Advice provided in a 2012 Biophysical Society profile and consistent with the "take-home messages" of her Duke exhibits.[4]
5. **On her career path and institutional standing**: "Well, I was certainly a real embarrassment to the university for a long time, once I became at all well known!... It was after I got into the National Academy of Sciences that Duke decided to give me tenure." *Context/Date*: Oral history/autobiographical reflection on the difficulties of not having a formal PhD while holding a professor position.[11]
6. **On the relationship between models and reality**: "The model is not the molecule." *Context/Date*: A core philosophical tenet of the 2020 exhibit "Seeing the Invisible: 50 Years of Macromolecular Visualization".[18]
7. **On the joy of the work**: "Science is fun, especially the details." *Context/Date*: Final takeaway from her 50-year retrospective exhibit.[18]

C) HUMAN REALITY DETAILS (8–15)

- **Partnership Longevity**: Jane and David Richardson have worked as a continuous research team since 1964, a rare 50+ year scientific collaboration.[1]

- **The Missing PhD**: Despite her massive influence, Jane Richardson never earned a PhD; she transitioned directly from a Master's in Philosophy to lab technician work.[1]
- **The Geodesic Virus Home**: Their residence in Durham is a geodesic dome mimicking an icosahedral virus, attached to an octagonal treehouse mimicking a T-even virus.[5]
- **High-Altitude Reflection**: Every summer, the Richardsons retreat to a remote cabin in the Sierra Nevada mountains to backpack for weeks, focusing on high-level reflection away from the lab.[14]
- **Manual Calibration**: In the early MIT days, David Richardson aligned their x-ray diffractometer by "gently tapping the base with a large hammer".[12]
- **DIY Lab Mentality**: They "reverse-engineered" protein crystallography by trial and error, testing growth conditions one at a time and writing all their own software.[1]
- **Stereo Intuition**: Jane can "visualize in her head the peptide orientations and H-bonding" simply by examining stereo images of a C_α -trace.[7]
- **Scientific "Gossip"**: Jane values conferences as a place to "gossip with people of all ages about science," emphasizing the human network of the biophysical community.[4]
- **Astronomy Hobbies**: Her fascination with structural observation began in third grade at the Hayden Planetarium; she built her own 6-inch telescope mirror and mount.[4]
- **The Leather Belt Experiment**: To master the drawing of twisted β -strands, she spent time holding and twisting a leather belt to observe how the light and edges interacted with the eye.[7]
- **Institutional "Outsider"**: Because she lacked a PhD, she was initially categorized as a "technical assistant" or "medical research assistant" for over 20 years before becoming a full professor.[9]
- **Free Laundry Tradition**: A quirky detail of their lab culture is that they have been known to "happily do your laundry for free" during collaborative stays.[1]

D) SCIENCE CONNECTION (INTRO BIOCHEMISTRY)

Best-fit topic:

Protein Structure: Secondary Motifs and Tertiary Folds.[11]

Precise contribution (1 sentence):

Jane Richardson established the "ribbon diagram" as the universal visual language for proteins and categorized the recurring topological patterns, such as the Greek key, that define how polypeptide chains fold into functional 3D shapes.[1]

Bridge explanation for intro students:

- **The Problem**: Proteins are composed of thousands of atoms, which, if drawn individually (space-filling models), look like an uninterpretable cloud of lumpy density.[4]

- **The Solution**: Richardson's diagrams "strip away" the atomistic noise to show the path of the backbone. This allows students to see α-helices (drawn as coils) and β-sheets (drawn as broad arrows).[11]
- **The Logic**: By focusing on the "skeleton" of the protein, students can see how hydrogen bonding patterns (secondary structure) organize into larger units like "Greek keys" or "barrels" (tertiary structure).[3]

Scientist's own framing/analogy:

Richardson compared her ribbon diagrams to **MC Escher's "rind" drawings**. Just as Escher uses a spiral strip to let you see both the "inside" and "outside" of a human face simultaneously, her ribbons allow a scientist to visualize the internal hydrophobic core and the external hydrophilic surface of a protein in a single glance.[7]

E) COMPLICATIONS, DISPUTES, OR CONTROVERSIES

Item #1: The Subjectivity vs. Algorithmic Control

- **What's agreed**: Richardson's original drawings were hand-drawn and involved artistic choices about which atoms to "average" into a smooth curve.[13]
- **What's disputed**: Some early peers argued with her about how she depicted specific molecules.[13] Today, the "danger" is that researchers trust computer-generated ribbons (e.g., in PyMOL) as objective truths, forgetting that the software is based on the subjective algorithms she pioneered.[11]
- **Sources**:.[11]

Item #2: Institutional Legitimacy and the "PhD Barrier"

- **What's agreed**: Richardson worked as a "technician" or "research assistant" for over 20 years despite doing world-class science.[9]
- **What's disputed**: She was considered a "real embarrassment" to Duke University's formal hierarchy until her election to the National Academy of Sciences essentially forced the university to grant her tenure.[11]
- **Sources**:.[9]

Item #3: Handedness of Greek Keys

- **What's agreed**: Greek keys are a fundamental topological signature of β-barrels.[16]
- **What's disputed**: There was an empirical claim that Greek keys in β-barrels followed a strict handedness regularity, but later analysis of representative structures challenged this, showing that half of the analyzed structures have left-handed Greek keys.[25]

- **Sources**:.[25]

F) “NOT ON WIKIPEDIA” GOLD (5+)

1. **The "Not Proteiny" Instinct**: Colleague Fred Brooks noted that Jane could look at a proposed computer model and simply say, "That's not proteiny," identifying errors through pure visual intuition that mathematical validation hadn't yet caught.[4]
2. **The Hammer Calibration**: David Richardson’s custom method for aligning the diffractometer involved "gently tapping the base with a large hammer"—a primitive bridge to the high-tech structural revolution.[12]
3. **MC Escher Influence**: The specific aesthetic of the ribbon diagram was directly inspired by Escher's "rind" drawings, representing a rare moment where 20th-century graphic art directly informed a structural biology standard.[7]
4. **The Fluorescent Tygon experiment**: Before drawing, the Richardsons used 1/4" tubing filled with dye and UV light on brass models to find the "worm" path of the nuclease structure.[7]
5. **The Virus-House Build Kit**: The Richardsons recognized halfway through building their home that the kit resembled a T-even virus and modified the plan to "fulfill" that biological shape.[14]
6. **The Paper Tape Era**: Their early crystallography data was captured using a machine controlled by punched paper tape, highlighting the mechanical-to-digital bridge in structural biology.[1]

G) SOURCES CONSULTED (Integrated Citations)

H) FLAGGED “MYTH-RISK” DETAILS

Claim: Jane Richardson is an "artist" who decided to apply her art to science. **Where it appears**: Various popular science blogs and casual profiles. **Why sourcing is weak**: Richardson herself explicitly states she "never considered herself an artist" and was "not being even an amateur artist" when she started the 1981 project.[7] **Best available evidence**: She describes her drawings as a result of "practicing how to draw" and "developing effective conventions" for scientific necessity rather than artistic expression.[7]

LENS RECOMMENDATION

Lens scoring rubric

Lens	Score (0–3)	Justification

Eureka	3	The "Tygon tubing" glowing light and the "Greek key" comparison are perfect, documented discovery moments.[7]
Failure	1	While the field was "hard" and they were "technicians," there isn't one specific failure that redefined her trajectory.
Mentor	2	Her relationship with Chris Anfinsen was pivotal; he suggested the tubing and commissioned the 1981 paper.[7]
Controversy	2	The institutional struggle over her lack of a PhD and "embarrassment" to Duke is well-documented.[11]
Human Detail	3	The geodesic virus house and Sierra Nevada backpacking trips are highly vivid and unique.[14]
Last Experiment	2	The SARS-CoV-2 validation work provides a strong "full circle" from drawing to digital validation.[22]

Primary recommended lens: Eureka (The Power of Visual Intuition)

- **Why it fits**: This lens connects her high-school astronomy (calculating an orbit from light points) to the Tygon tubing (seeing the backbone through UV light) and finally to the 1981 taxonomy (the year of drawing). It matches her "not proteiny" instinct.[1]
- **Best opening scene**: Scene #1 (Sputnik). It establishes her as an observer who trusts her own data.
- **One quote to use early**: Quote #1: “I think my philosophy background is actually useful, mainly in questioning absolutely everything.”
- **Cautions**: Avoid overstating the "artist" angle (see Myth Risk). Focus on the technical

rigor of her "practicing" the drawings.[7]

Backup lens: Controversy (The Credential Barrier)

- **Why it fits**: This centers the narrative on the institutional struggle of being the "Mother of Ribbon Diagrams" while being officially titled a "technician." It highlights the gender and degree politics of mid-20th-century science.[11]
- **Best opening scene**: Scene #4 (The Marathon Drawing).
- **One quote to use early**: Quote #5: "Well, I was certainly a real embarrassment to the university...".[11]

Note: This report is provided to the student as raw material. Do not write the finished profile.

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Profile

On a biting night in Teaneck, New Jersey, seventeen-year-old **Jane Richardson** stands in the cold, her eyes fixed on the celestial void[1]. She is not merely stargazing; she is hunting for the man-made light of Sputnik I as it streaks across the sky[2]. Between counts of passing meteors, she adjusts a six-inch telescope mirror—an object she ground and polished with her own hands[3]. She records the coordinates of the satellite, translating fleeting points of light into mathematical predictions of an orbit[4]. This is the labor of a rigorous observer, someone capable of extracting high-level patterns from raw, complex data long before she ever encounters a protein[5].

The stakes of such observation become even higher years later at MIT, where the data is no longer astronomical but molecular[6]. Richardson and her husband, David, are staring at a brass wire model of Staphylococcal nuclease, only the tenth protein structure ever solved[7]. The model is a "dense forest" of wires representing thousands of atoms, a chaotic thicket that obscures the very architecture it is meant to reveal[8]. "I think my philosophy background is actually useful, mainly in questioning absolutely everything," Richardson notes of her unconventional path from a Master's degree in philosophy to a laboratory bench[9].

The breakthrough arrives not through a computer, but through a DIY experiment involving 1/4-inch Tygon tubing[10]. Following a suggestion from colleague Chris Anfinsen, the Richardsons thread the flexible tubing along the protein's backbone and fill it with fluorescent dye[11]. When they switch on a UV light, the "forest" of individual atoms vanishes into the shadows[12]. In its place, the "worm-like" path of the backbone emerges as a single, glowing, continuous line[13]. It is a literal lightbulb moment: to understand the whole, one must have the courage to omit the details[14].

This visual intuition allows Richardson to see proteins as architectural forms rather than "frozen accidents" of nature[15]. She spends an entire year hand-drawing every

known protein structure, holding up a twisted leather belt to her eyes to study how the edges of a surface should curve in three dimensions[16]. Most proteins are composed of thousands of atoms which, if drawn individually, look like an uninterpretable cloud of lumpy density[17]. Richardson's ribbon diagrams strip away this noise to show the path of the backbone, using coils to represent α-helices and broad arrows for β-sheets[18]. This skeleton reveals how the protein folds into functional 3D shapes, such as the interlocking "Greek key" motif[19].

Decades later, during the 2020 global pandemic, Richardson sits at a screen in North Carolina, applying this same visual rigor to the flood of new SARS-CoV-2 protein models[20]. She and her team work behind the scenes to flag "peptide flips" and misalignments that mathematical validation alone might miss[21]. It is a digital echo of that cold night in Teaneck. Whether tracking a satellite through a hand-ground mirror or a virus through a software algorithm, she remains the master of the "not proteiny" instinct—the observer who knows that the model is not the molecule, but a necessary map for seeing the invisible[22].